

FedDG-MoE-NF

Prototypical Normalizing Flow Networks for Federated Domain Generalization

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CODE

THE PROBLEM

Federated Domain Generalization trains on decentralized source domains and is then deployed on a target domain **no client has ever seen**.

Vision Transformers are the strongest available feature extractors, but tuning them per client is prohibitive: millions of parameters cross the network every round, and scarce local data invites overfitting. Parameter-efficient adapters and Mixture-of-Experts fix the cost — they do not answer the harder question.

Given one expert per source client, **how do you combine them for a domain you have never observed?**

Existing methods answer with first-order statistics — a running **feature mean** per domain. In data-scarce regimes a mean vector cannot separate distributions differing in shape, spread or multimodality, so adaptation runs on an impoverished description of each domain.

THREE KINDS OF SHIFT

Each benchmark is split leave-one-domain-out: three domains become federated clients, the fourth is held back as the unseen target.

OfficeHome 15.5k · 65 cls	Art, Clipart, Product, Real-World — a test of style invariance .
PACS 9.9k · 7 cls	Photo, Art, Cartoon, Sketch — severe texture loss demands shape bias .
VLCS 10.7k · 5 cls	VOC, LabelMe, Caltech, SUN — all photographic, so the shift is capture bias .

OUR ANSWER

- **Frozen ViT + Kronecker MoE adapters.** High-rank adaptation from strictly low-rank parts, with only a fraction of parameters trainable per client.
- **Densities, not means.** Each client fits a Normalizing Flow to its own features, giving *exact likelihoods* instead of a summary statistic.
- **Density-guided test-time fusion.** Two new strategies — prototypical similarity and likelihood weighting — compose experts on the fly for the unseen domain.
- **State of the art on three benchmarks.** OfficeHome, PACS and VLCS, against 15 centralized, federated and PEFT baselines.

1 MIXTURE OF KRONECKER EXPERTS

The FFN of each ViT block is replaced by n experts. Every expert approximates its weight update as a Kronecker product, which builds a **high-rank** update out of two small factors:

$$\Delta W = \sum_{i=1}^n A_i \otimes B_i, \quad B_i = u_i v_i$$

with $d_{\text{out}} = p \cdot r$, $d_{\text{in}} = q \cdot s$, and $u_i \in \mathbb{R}^{r \times k}$, $v_i \in \mathbb{R}^{k \times s}$ factorizing each expert further.

Slow weights A_i

Shared across clients and aggregated by the server — the universal, domain-agnostic part.

Fast weights $B_i = u_i v_i$

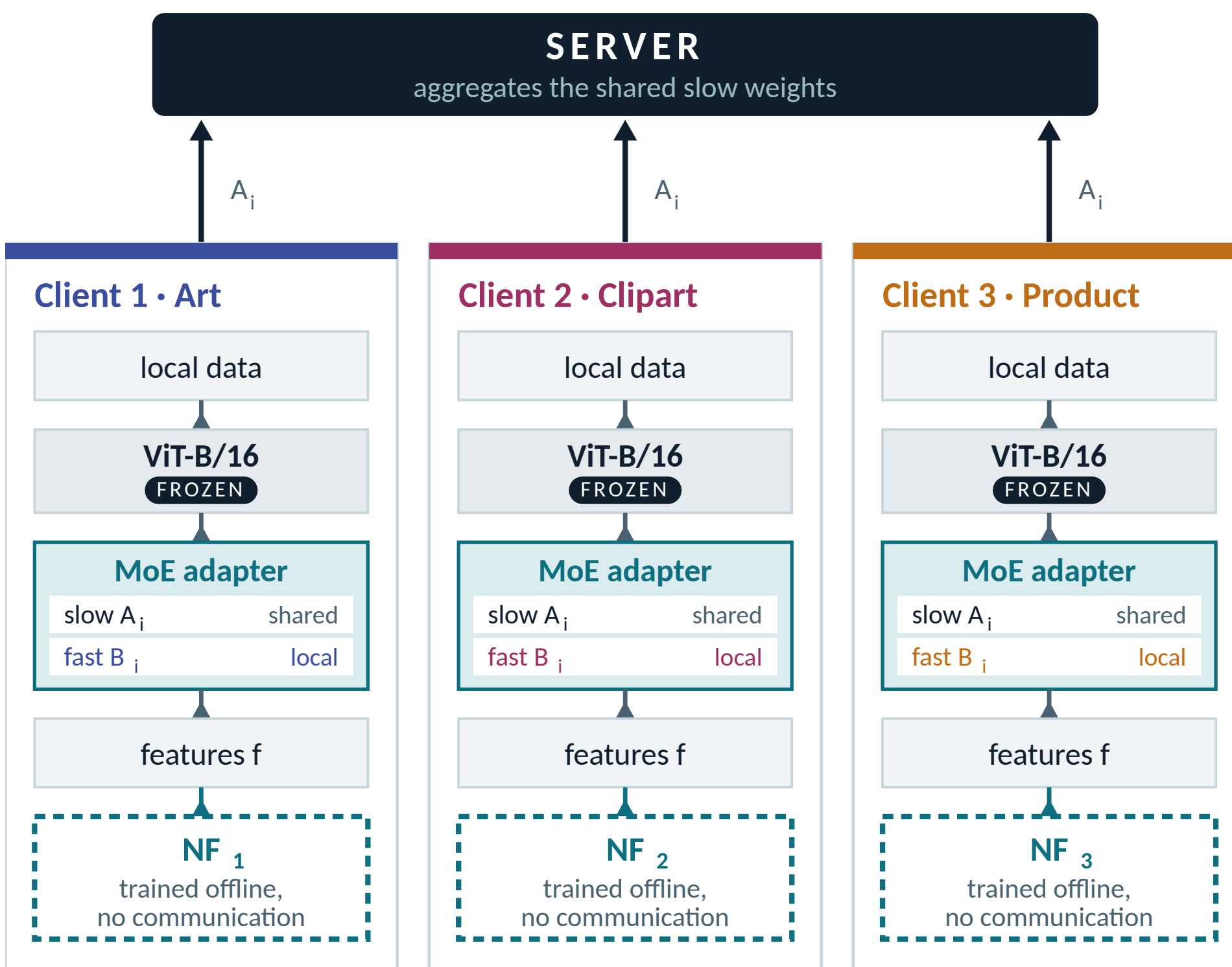
Client-specific and never transmitted — they carry whatever is unique to that domain.

A lightweight router turns each input into gating coefficients α_i , so different inputs recruit different expert combinations:

$$y = \sum_{i=1}^n \alpha_i (A_i \otimes (u_i v_i)) x_{\text{feat}} + b$$

Only the adapters and the head receive gradients. The 86M parameters of the CLIP-pretrained ViT-B/16 stay fixed and never cross the network — that is where the communication saving comes from.

FEDERATED TRAINING



Only the adapters and the classification head are trained. Slow weights are aggregated at the server; fast weights, flows and raw data never leave the client. Shown for OfficeHome, Real-World held out.

2 DENSITY MODELLING WITH FLOWS

Each client k then fits a flow NF_k — a bijection carrying its empirical feature density $p_k(f)$ onto a standard Gaussian — by minimizing the negative log-likelihood:

$$\mathcal{L}_{\text{NF}_k} = \mathbb{E}_{f \sim \mathcal{D}_k} \left[\frac{1}{2} \|\text{NF}_k(f)\|_2^2 - \log |\det J_{\text{NF}_k}(f)| \right]$$

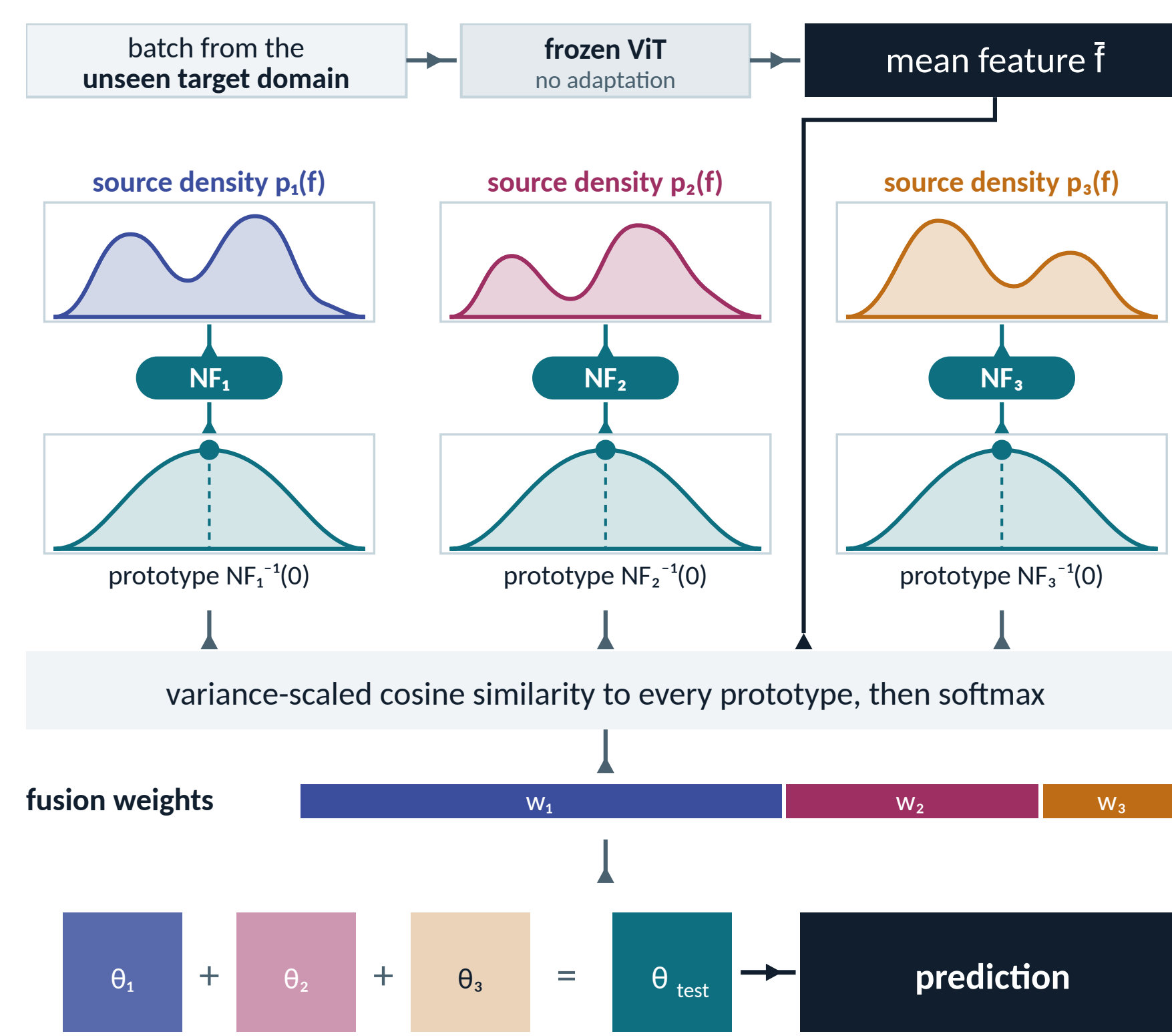
The result is an **exact, differentiable description of the domain** rather than a summary statistic — and it is fitted entirely offline, adding nothing to the communication budget.

Flow configuration we settled on

RealNVP hidden dim **128** **30** epochs weight decay **10^{-5}**

Smaller and shorter wins: on OfficeHome, hidden dim 768 costs 0.47 points and 60 epochs 0.56 — both consistent with overfitting on limited client data. GLOW (88.57) and GIN (88.75) trail RealNVP (88.96).

3 FUSING EXPERTS AT TEST TIME



Inference on an unseen domain. Colours match the source clients above. Each flow contributes a prototype — the feature mapping to the mode of its latent Gaussian — and its variance-scaled similarity to the test batch sets the weights that mix the client adapters.

Prototypical similarity — best

$$f_k^{\text{proto}} = \text{NF}_k^{-1}(0), \quad s_k = \cos \left(\frac{\bar{f}_{\text{test}}}{\sigma_k + \epsilon}, \frac{f_k^{\text{proto}}}{\sigma_k + \epsilon} \right)$$

$$w = \text{softmax}(s/T), \quad \theta_{\text{test}} = \sum_k w_k \theta_k^{\text{MoE}}$$

Likelihood variants

$$\mathcal{L}_k = \frac{1}{|B_{\text{test}}|} \sum_{x \in B_{\text{test}}} \left[-\frac{1}{2} \|\text{NF}_k(f(x))\|_2^2 + \log |\det J_{\text{NF}_k}| \right]$$

Used three ways — soft weighting, hard selection of $\arg \max_k \mathcal{L}_k$, or sampling from the softmax. Appealing in theory, less stable in high dimensions; see the ablation.

RESULTS

OFFICEHOME 88.96% +5.37 vs. PromptFL	PACS 97.40% +0.76 vs. PromptFL	VLCS 83.35% +1.52 vs. FedCLIP
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Leave-one-domain-out average accuracy. Frozen CLIP ViT-B/16 backbone; Adam, $\text{lr } 10^{-3}$, batch 64, 5 communication rounds $\times$ 10 local epochs; fusion temperature $T = 0.5$.

OFFICEHOME · LEAVE-ONE-DOMAIN-OUT (%)

Method	P	A	C	R	Avg
CENTRALIZED					
Doprompt	88.94	80.95	70.88	90.10	82.72
FEDERATED, FULL FINE-TUNING					
FedAvg (ViT-B)	80.15	62.72	71.11	81.13	73.78
FedAvg (ConvNeXt-B)	89.81	73.31	84.42	90.31	84.46
ELCFS + GA	73.35	68.62	50.60	81.23	68.45
PARAMETER-EFFICIENT (ViT-CLIP)					
FedCLIP	87.68	78.45	64.77	87.84	79.69
PromptFL	92.14	82.98	68.98	90.27	83.59

OURS

FedDG-MoE (AM)	94.84	85.33	82.70	92.56	88.86
FedDG-MoE-NF	94.95	85.46	82.02	93.41	88.96

Held-out domain: P Product, A Art, C Clipart, R Real-World. Representative baselines shown; the paper compares 15 methods. The FedDG-MoE row isolates the adapter design — NF adds density-guided fusion on top.

PACS & VLCS

PACS

Doprompt centralized	89.91
FedAvg ConvNeXt-B	92.55
FedCLIP	94.93
PromptFL	96.64
FedDG-MoE-NF	97.40

Axis 78–100. Best per-domain: 99.94 Photo, 93.23 Sketch.

VLCS

FedAvg	78.16
HCVP centralized	81.08
PromptFL	81.26
FedCLIP	81.83
FedDG-MoE-NF	83.35

Axis 75.7–86. Per-domain: 85.39 VOC2007, 100.00 Caltech-101.

WHICH FUSION RULE?

FUSION RULE, OFFICEHOME (COLUMNS AS ABOVE)

Fusion	P	A	C	R	Avg
NF prototype cosine	94.95	85.46	82.02	93.41	88.96
Hard selection	94.12	83.23	80.87	91.78	87.50
Likelihood weighting	94.30	83.19	80.73	91.51	87.43
Probabilistic selection	94.30	82.90	80.57	91.58	87.34

Matching the test batch to **simplified prototypes** beats raw likelihoods by 1.5 points: in high dimensions, likelihood estimates are dominated by outliers, while the prototype is a stable point of comparison.

WHAT TO TAKE AWAY

- A frozen backbone plus Kronecker MoE adapters is enough to beat full fine-tuning — at a small fraction of the communication cost.
- Modelling a client's feature , not its mean, gives test-time fusion something real to work with.
- The flows are free at training time: fitted offline, per client, with no extra rounds.

Next: dynamic expert allocation, and density-guided fusion for multimodal federated learning.

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